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Soil macroporosity in relation to subsurface drain location on a sloping clay field in humid climatic conditions

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ABSTRACT

Macroporosity of soil and earthworm abundance in relation to the distance from subsurface drain lines were determined to investigate the effect of subsurface drainage on the spatial variation of clay soil structure. Our hypothesis was that soil macroporosity near a drain line location would be greater than the macroporosity at a midpoint between two drain lines where the groundwater table remains theoretically at a higher level. The experimental site was located on a post-glacial clay soil classified as a very fine Aeric Cryaquept (Vertic Cambisol). The area of the field was 1 ha and the slope varied from 2 to 4%. Measurements of soil properties were carried out along four subsurface drain lines installed 50 years before soil sampling. In each drain line, vertical soil profiles were examined at different distances from the drain: (1) above the drain, (2) 2 m from the drain, and (3) midway between two drains (6–7 m from a drain). In each vertical profile, soil macroporosity, number of earthworm burrows and root channels, and saturated hydraulic conductivity (K_{sat}) were determined in three soil layers to the depth of 50 cm. In addition, the field saturated hydraulic conductivity (*in situ*) and the number and fresh mass of earthworms were determined.

The results showed that the number of earthworm burrows was 44% (90 burrows m^{-2}) greater above a drain than at a distance of 6–7 m from the drain. There was also an indication of higher earthworm density above the subsurface drains than between them.

The other measures of soil physical properties were not clearly related to distance from the drain. One reason for the small differences in soil properties in relation to drain position was probably the groundwater level that in our clay field varied less in relation to drain location than was expected. Spatial variation of the investigated soil properties was mostly explained by the slope of the field. Even though the groundwater table was shallower at the downslope part compared with the upslope part of the field, the number of biopores, K_{sat} and the number of earthworms clearly increased downslope.

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1. Introduction

Soil structure, i.e. porosity and aggregation, markedly affects the cultivability and workability of clay soils. The number, continuity and spatial distribution of soil macropores significantly affect the hydrological, chemical and biological processes in clay soils by moderating the rate of water infiltration and soil drying after rainfall or snow melting (Beven and German, 1982; Bouma, 1991; Aura, 1995). Formation and the stability of clay aggregates are important factors for the reduction of slaking, the generation of

surface runoff, the erodibility of soil and the transport of particulate phosphorus in clay fields (e.g. Rhoton et al., 2002).

Soil drying is an essential physical process influencing the structure development of clay soil. Through making the water film thinner around the soil particles, soil drying increases the contacts between particles and enhances aggregate formation (van de Graaff, 1979). Moreover, unsaturated soil moisture conditions favour shrinking and cracking, root growth, and the activity of soil organisms. A number of studies have shown that the physical properties of clayey soils with respect to crop production characteristics improve when the soil moisture content decreased (Caron et al., 1992; Hermawan and Bomke, 1996).

Lowering of the groundwater table by subsurface drainage decreases soil moisture in the unsaturated zone and enhances physical and biological processes that improve the structure of clayey soils. van de Graaff (1979) reported that poorly drained soils were generally wetter and dried more slowly, less severely, and

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less frequently than well drained soils. As a consequence, there was less intensive cracking in the poorly drained soils. An example of biological factor with important implications for the formation of soil structure is earthworm activity, and this is enhanced when wetlands and water-logged soils are drained (van Rhee, 1969; Carter et al., 1982; Baker, 1998). Subsurface drainage has been reported to improve the aggregation and aggregate stability of clay soil (Baker et al., 2004), and to increase the volume of air-filled pores (Hundal et al., 1976) and saturated hydraulic conductivity (Bouma et al., 1979).

Finland's most important agricultural region for annual crops is the southern and southwestern part of the country, located between latitudes 60° and 61°N. The mineral soils in southern Finland are young with respect to geological age, mainly deposited by glacial activity 9000–12,000 years B.P. (Donner, 1995). Clay soils typically contain more than 30 g 100 g⁻¹ clay (diameter < 2 µm). The clay minerals are mostly of illitic type (Sippola, 1974). Clay subsoil characterizes more than 60% of the arable land in south and southwestern Finland and 35% of the arable land (2.3×10^6 ha) in Finland as a whole (Puustinen et al., 1994).

Subsurface drainage, consisting of perforated pipes or tiles installed at depths typically ranging from 70 to 100 cm, is the most common method of water table management for clayey soils in Finland. In the southern and southwestern parts of the country, 75% of the arable area is subsurface drained (Yearbook of Farm Statistics, 2003). The lowering of the water table by subsurface drainage has markedly affected the structure development of Finnish clay soils (Mokma et al., 2000). Subsurface drained soils dry up to a greater depth than soils with natural water table, and this is presumed to cause irreversible changes in the soil structure.

Even though drawing down the groundwater table by subsurface drainage is expected to be beneficial for the structure development of clay soils, the effect of drainage on the macroporosity of soil has not been widely examined. Modifications in soil macroporosity are very important since they clearly affect several physical, chemical and biological soil properties and processes which may have marked economic and environmental consequences. There is a wide interest towards exploring the effects of subsurface drainage on soil structure and water movement, because nearly all the agricultural fields in our study region are subsurface drained.

We addressed the question by studying the relationship between distance from the subsurface drain line and the variation of macroporosity in clay soil under humid climatic conditions. A further aim was to interpret how macroporosity influences the

generation of subsurface drain flow (Hintikka et al., 2008). The studied subsurface drainage system, located in southern Finland, was installed in 1951. Clear evidence of preferential water flow and nutrient transport via the subsurface drainage system had been observed in the present field (Paasonen-Kivekäs et al., 1999; Uusitalo et al., 2001). The preferential flowpaths were assumed to occur in natural macropores (biopores and cracks) and in the backfill material of the drainage trenches. However, no direct measurements of macroporosity or spatial variability in the vicinity of the drainage trench have been available from this field thus far.

We expected that soil macroporosity near the drain location would be greater than the macroporosity at locations between two drain lines where the groundwater table is, according to the theory of drainage, at higher level (Smedema and Rycroft, 1983). We based our hypothesis partly on earlier observations on influence of clay soil structure on crop growth and field saturated hydraulic conductivity (K_{fsat}). On clay soil in rainy years, crop growth near subsurface drain lines has been noticed to be better than between drain lines. Likewise, Shipitalo et al. (2004) found higher K_{fsat} above drain compared to midpoint between drains in a clay field in southwestern Finland.

2. Materials and methods

2.1. Experimental site and sampling locations

The Sjöckulla experimental field was instrumented for non-point source pollution studies by the Helsinki University of Technology in 1992–1994. The site is located in Kirkkonummi in the southern part of Finland (60°15'N, 24°27'E, Fig. 1). The soil in the area is clay and classified (Soil Survey Staff, 1998) as a very fine Aeric Cryaquept (Peltovuori et al., 2002), and according to FAO classification as a Vertic Cambisol (FAO, 1988). The topmost layer down to a depth of approximately 30 cm comprises freshwater sediments underlain by brackish water sediments (Peltovuori et al., 2002). The topmost clay sediments in the area were formed after the last ice-age in Scandinavia. The soil was exposed from the water about 7500–8000 years ago (Uusitalo et al., 2001). The Sjöckulla field has likely been under cultivation since the sixteenth century. Some characteristics of the soil at different locations on the field are given in Fig. 2. The main clay mineral was illitic (Peltovuori et al., 2002). Average soil pH (water, 1:2.5) was 6.3 at the topsoil (0–25 cm) and 7.5 in the deepest soil layer (65–80 cm). When the soil dried, it shrank noticeably and, during the

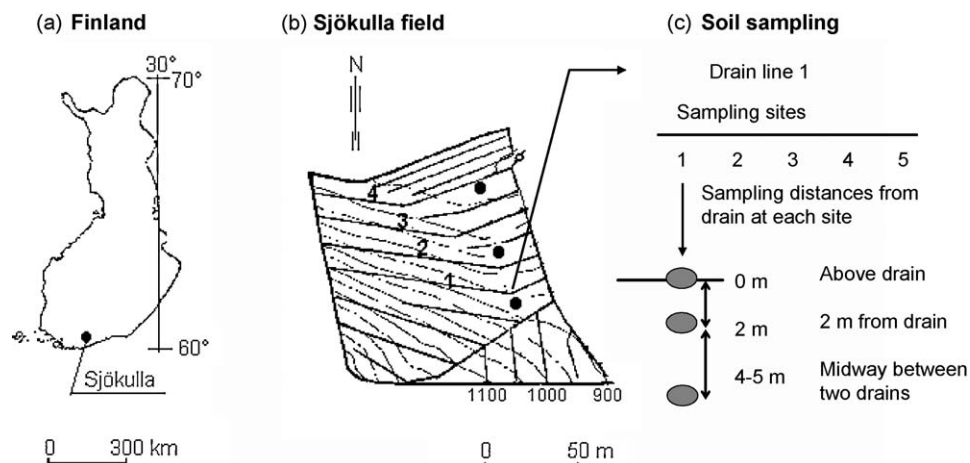


Fig. 1. (a) Location of the Sjöckulla experimental field and (b) layout of the subsurface drainage network. Measurements were conducted along drain lines 1–4 at three to five sites. At each site (c) there were three sampling distances from the drain. Drain line 1 was located at the upper part of the slope and drain line 4 at the bottom part of the slope. Circles (●) show the location of groundwater observation tubes.

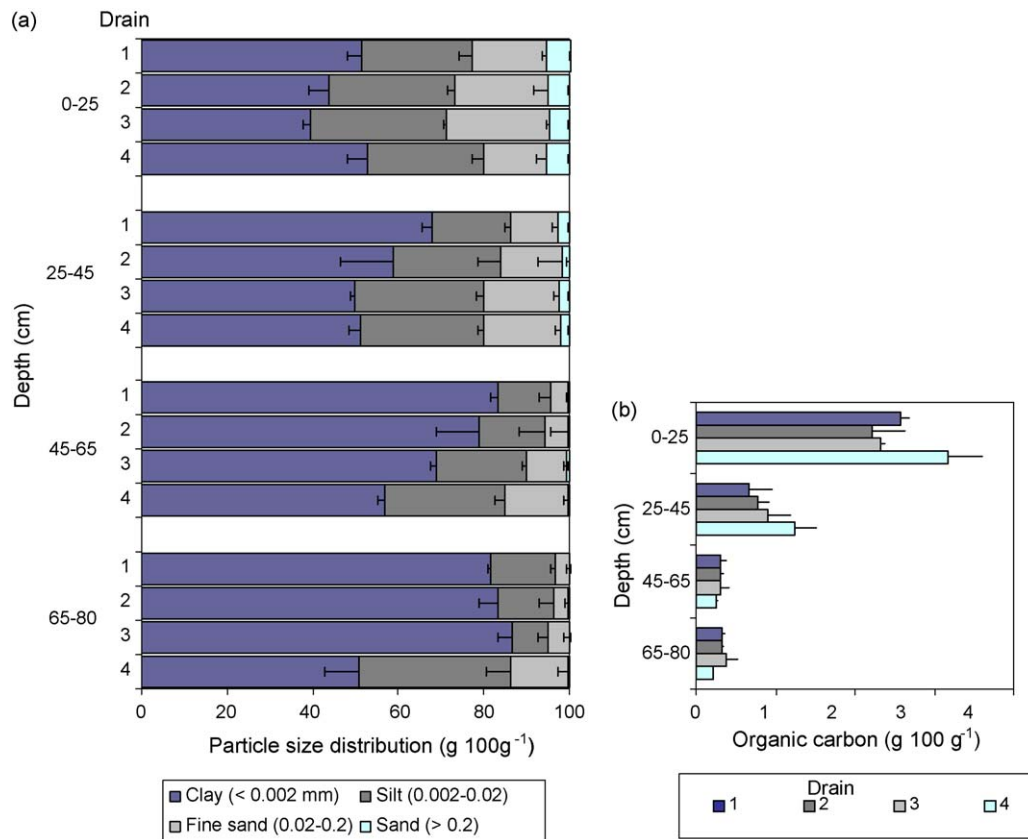


Fig. 2. (a) Average soil particle size distribution and (b) organic carbon content as a function of soil depth at midpoint between two subsurface drains (Fig. 1). The values are medians of 3–5 measurements, and bars denote quartile deviations.

experimental period in 2001, cracking was observed at least down to a depth of 60 cm.

The study site was a sloping field section with an area of 1 ha. The steepest/main hillslope had a length of 140 m and a slope of 2–4%. The field was subsurface drained in 1951 when a narrow trench (about 20 cm wide) was dug and drain tiles were installed at the bottom of the trench. The tiles were coated with a thin layer of gravel before the trench was back filled with the original soil. The average depth to drains was 110 cm and the drains were spaced about 13 m apart. Spring cereals were the main crop during the 10 years preceding the soil sampling in 2001. Since autumn 1999, the field had been under stubble cultivation (working depth 15 cm). Before 1999, the field was autumn ploughed to a depth of 23 cm (years 1992, 1996, 1998) or stubble cultivated (years 1993–1995, 1997).

Before the field study, subsurface drains were located by pushing a metal spike (with a cone) through the soil against the tile. During the sampling in September–October 2001, soil moisture content was near field capacity and the soil was swollen after a rainy period. These soil conditions are typical at times when spring snowmelt or autumn rainfalls lead to a generation of high runoff volumes. Soil samples were taken and earthworms were sampled along four subsurface drain lines (Fig. 1). Each subsurface drain line included three to five sampling sites (Fig. 1). The distance between two sites along the same line was 10–15 m. In total, 16 sites were sampled. Earthworms were sampled, and field saturated hydraulic conductivity was measured at a distance of 1–2 m from the sampling sites. At each site, samples were taken at two distances from the drain line: (1) above the drain and (2) midway between two drains (6–7 m from a drain). Additional soil samples were taken (3) at a distance of 2 m from each drain line.

Fig. 3a and b shows average soil moisture content at two depth ranges (0–30 and 0–60 cm) at locations above the drains and between two drains during the growing season 2001. Depths to groundwater table at midway between two subsurface drains are also plotted (Fig. 3c). Soil moisture content was measured with a TRASE-TDR (Soil Moisture Equipment Corp., CA, USA). The groundwater level was determined by using bottom closed, perforated, plastic groundwater observation pipes.

Climate in the Sjöckulla area is temperate with mean annual precipitation of 700 mm and temperature of +5 °C. During the measurement period in 2001, the monthly precipitation was clearly higher (122 mm in June and 99 mm in September) than the long-term average values (49 and 69 mm, respectively).

2.2. Measurements of soil physical properties

Field saturated hydraulic conductivity (K_{fsat}) was determined with a 0.5 m² (A) single ring infiltrometer (Smedema and Rycroft, 1983; Aura, 1990). The infiltrometer ring was installed to a depth of 10 cm (z , corresponding here the profile height), and before the measurements topsoil was moistened to field capacity. During the measurement a constant head of 10 cm of water (h) above the soil surface was maintained in the ring by a Mariotte siphon, and volume of infiltrated water (V) was determined at 2-min intervals (t , time) to determine the infiltration rate (q). When the infiltration rate had reached steady state (q_r), this was used to estimate K_{fsat} according to Darcy's law (Smedema and Rycroft, 1983):

$$q_r = \frac{V}{A \cdot T} = \frac{K_{fsat}(h+z)}{z} \quad (1)$$

where $(h+z)/z$ is the average hydraulic gradient.

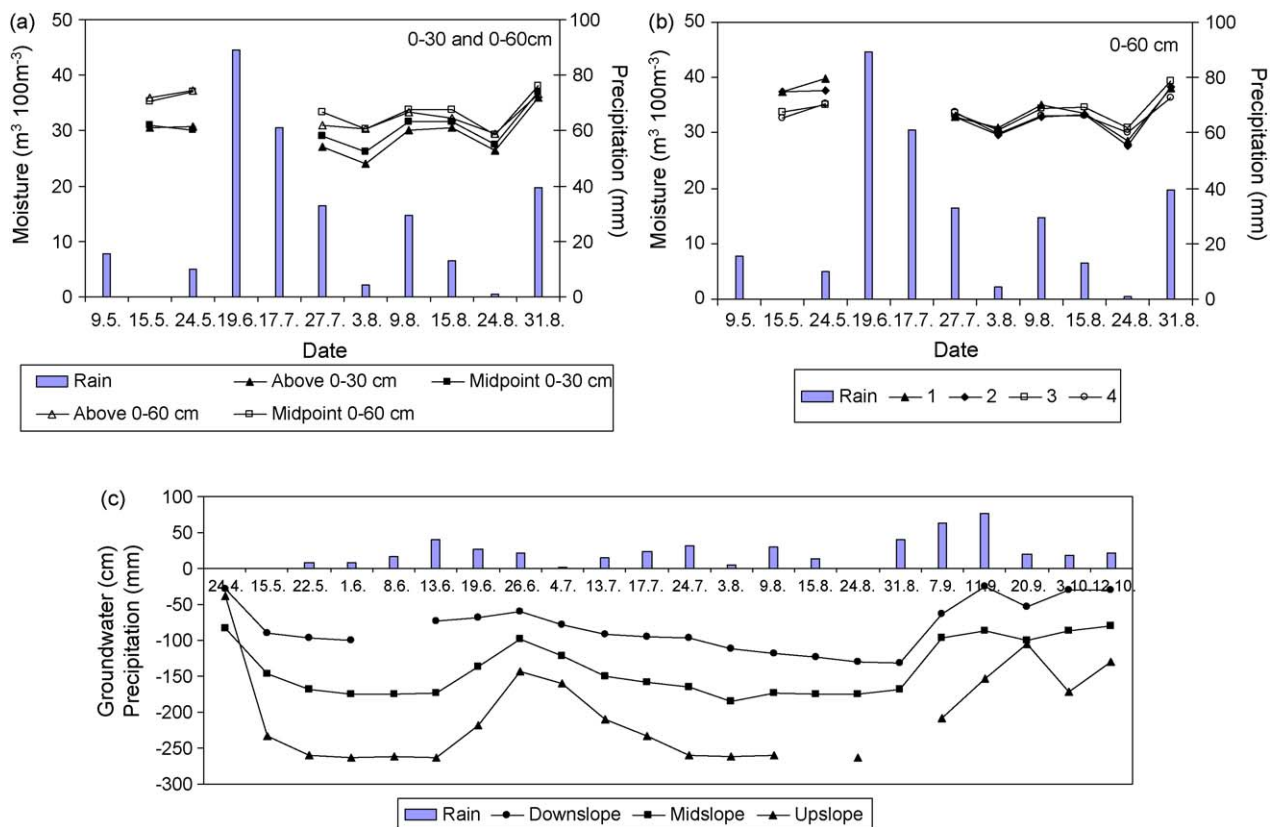


Fig. 3. (a) Average soil moisture content at two different depth ranges (0–30 and 0–60 cm) and two different locations (above subsurface drains and at the midpoint between two drains), (b) mean soil moisture content at a depth range from 0 to 60 cm midway between drains for drain lines 1–4, and (c) depth to groundwater table from the soil surface at midpoint between subsurface drains in 2001. The bars in (a)–(c) indicate precipitation after the previous measurement date.

In the cylinder the water the flow was one-dimensional but below the bottom of cylinder the flow was evidently transformed to a three-dimensional flow. Thus, the real infiltration rate for vertical flow was probably lower than inside the infiltrometer ring determined infiltration rate. The depth of the wetting front increased during the measurement being greater than z used to calculate the K_{fsat} . Even though the lateral movement of water and the lack of information about exact depth of wetting front reduced the accuracy of absolute values, the determination gave comparable relative values of K_{fsat} for treatments.

Undisturbed soil samples (PVC pipe 15 cm in diameter, 55 cm long) for laboratory analysis were taken with a tractor-driven soil auger at an undisturbed position next to the site of the K_{fsat} measurement. Before analysis the samples were stored at 5 °C. The samples were cut into three subsamples: plough layer (0–23 cm), middle layer (15 cm below plough layer) and bottom layer (the rest of the sample). To obtain a surface with open macropores, cut surfaces were prepared by removing any smeared or damaged soil with a knife and vacuum cleaner. First the number of cylindrical pores ≥ 2 mm (earthworm burrows) in the prepared surfaces was counted, and then the number of cylindrical pores < 2 mm (mostly root channels) was determined by counting the number of pores from five separate 2 cm $\times$ 2 cm areas of the surface. Earthworm burrows and root channels together were considered as biopores.

The subsamples were saturated with boiled water from bottom to upwards to avoid air entrapment. Samples were then soaked for five days and saturated hydraulic conductivity (K_{sat} , constant head method; Youngs, 1991) of soil was determined. Soil macroporosity (theoretical diameter > 300 μ m, largest macropores relevant to preferential flow) of the subsamples was determined by measuring the volume of water out flowed from a saturated soil sample at -1

to -0.75 kPa water potential (determined from the centre of the sample height) as described by Alakukku (1996). The subsamples were then put on ceramic plates (air entry value 100 kPa) and the water potential was equilibrated to bottom level of the sample. After that the macroporosity measurement at a potential of -10 kPa, corresponding to a minimum pore diameter of 30–300 μ m, was carried out by water hanging method as reported in Aura (1995). The pore diameter (μ m) corresponding to the water potential was estimated by dividing 3000 by the absolute value of water potential as height of water column (cm) (e.g. Czeratzki, 1958).

2.3. Earthworm abundance

In earthworm sampling, chemical extraction with mustard vermifuge (Gunn, 1992) was first applied in a 0.5 m² area delimited with an infiltrometer ring. Then a soil sample (20 cm $\times$ 20 cm $\times$ 15 cm) was taken from the middle of the ring area for hand-sorting of worms. Sampling was carried out for 15 sites located above a drain line and for 15 sites between two drain lines. Additionally, chemical extraction alone was applied at 2 m distance from the drain for a detailed analysis of the horizontal distribution of the deep burrowing *Lumbricus terrestris*. During sampling, the average moisture content in the topsoil layer (0–15 cm) was 36 m³ 100 m⁻³ and the temperature near the soil surface (0–10 cm) varied between 7 and 8 °C. Earthworms stored in 70% alcohol were identified according to Sims and Gerard (1999), and the fresh mass of each individual was determined. In the case of damaged, incomplete specimens, only heads were counted as individuals. Estimates of abundance per square metre were calculated by multiplying the numbers of worms in

chemically extracted and hand-sorted samples by the inverse of the corresponding sample area and adding the figures together.

2.4. Statistical analyses

In analysing how the distance from a drain line influences soil physical properties and earthworm abundance, mean values of the measurements at the three to five sampling sites along each drain were used as observations. Analyses of the total number and fresh mass of earthworms measured at two distances (0 and 6–7 m from drain line) were based on the following model:

$$y_{ij} = \mu + \delta_i + L_j + \varepsilon_{ij}, \quad (2)$$

where y_{ij} denotes the observation for the drain i at distance j , μ is a constant, δ_i is the random effect for drain i , L_j is the fixed effect for distance j , and ε_{ij} is the random error. The random effects δ_i and ε_{ij} are assumed to be independent and normally distributed with zero means and variances σ_δ^2 and σ_ε^2 , respectively. Furthermore, δ_i and ε_{ij} are assumed to be independent of one another. The assumptions imply that two observations within range of the same drain, one above the drain and the other 6–7 m from the drain, are intercorrelated, i.e. the covariance between the observations (σ_{12}) deviates from zero. The assumptions also imply that the variances of observations at the two distances are equal ($\sigma_1^2 = \sigma_2^2 = \sigma^2$). Consequently, the variance–covariance matrix Σ for the observations within a drain is of the form

$$\Sigma = \begin{bmatrix} \sigma^2 & \sigma_{12} \\ \sigma_{12} & \sigma^2 \end{bmatrix} \quad (3)$$

Field saturated hydraulic conductivity (K_{fsat}) was measured at three distances from a subsurface drain and the covariance structure used for the observations was of the following “unstructured” form:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 \end{bmatrix} \quad (4)$$

In this structure the variances and the pairwise covariances of observations at the three distances are allowed to be unequal. The values of all the other response variables were measured at the three distances from a drain line as well as at three depths within each distance. The model for observations of soil macroporosity (diameter > 300 μm) was the following:

$$y_{ijk} = \mu + \delta_i + L_j + D_k + LD_{jk} + \delta L_{ij} + \delta D_{ik} + \varepsilon_{ijk}, \quad (5)$$

where y_{ijk} denotes the observation for the drain i at distance j and at depth k . μ , δ_i and L_j are as in model (1); D_k and LD_{jk} are the fixed effects

for depth and distance-by-depth interaction, respectively; δL_{ij} and δD_{ik} are the random effects representing interactions between the random drain effect and the effects of distance (L) and depth (D); and ε_{ijk} is the random error. The random effects δ_i , δL_{ij} , δD_{ik} and ε_{ijk} are assumed to be independent, and within each term the effects are assumed to be independent and identically normally distributed with zero means. The assumptions imply for both distance and depth a compound symmetry covariance structure in which the pairwise covariances are equal and, similarly the variances of observations at different distances or depths are equal. This structure also fitted adequately the saturated hydraulic conductivity data measured in the laboratory (K_{sat}) and the data for the number of earthworm burrows after square root transformation for both variables. In the analysis of soil macroporosity (diameter > 30 μm), the “unstructured” covariance matrix (see above) appeared appropriate for both distance and depth. In the “best” models for the number of root channels and area of earthworm burrows (both square root-transformed), the covariance structures were “unstructured” and compound symmetry for distance and depth, respectively.

The models were fitted by using the method of Residual Maximum Likelihood (REML). The covariance structures were chosen through a comparison of different models by the likelihood ratio test (Littell et al., 1996). Furthermore, the adequacy of the models was examined through Akaike's and Schwarz's criteria (Littell et al., 1996) and the graphic analysis of residuals. The normality assumption of the residuals was checked by using the box plot (Tukey, 1977), and if necessary, appropriate transformations of response variables were used to obtain normality. The residuals were also plotted against the fitted values. The significances of the effects were determined by F -tests. Comparisons between estimated group means were made by two-sided t -type tests. The accuracies of the mean estimates were expressed through standard errors (SE) and, when transformations were used, through 95% confidence intervals (CI). The analyses were performed by the MIXED procedure in SAS/STAT software (Littell et al., 1996). Median and quartile deviations (quartile range divided by 2) were used as descriptive statistics because part of the outcome variables were not normally distributed.

3. Results and discussion

3.1. Macro- and bioporosity

The mean values of soil macroporosity clearly decreased from the depth of 23 cm to the depth of 50 cm but did not differ substantially at the three distances from the subsurface drain (Table 1). Also, the differences in the mean number of root channels between distances were small ($P = 0.95$, Table 2 and

Table 1

Model-based mean estimates of soil macroporosity at three different distances from a subsurface drain and at three different sampling depths. SE is the standard error of the mean. F is the value of F -statistic for the main effects of distance and depth with degrees of freedom as subscripts (numerator, denominator).

Macroporosity > 300 μm^* ($\text{m}^3 100 \text{ m}^{-3}$)			$F_{2,6}$	P -value	Macroporosity > 30 μm ($\text{m}^3 100 \text{ m}^{-3}$)		$F_{2,6}$	P -value
Mean	SE				Mean	SE		
Distance from drain (m)			0.56	0.60			0.09	0.92
0	1.2	0.1			3.0	0.5		
2	1.4	0.1			3.2	0.2		
6–7 m	1.2	0.1			3.1	0.2		
Depth (cm)			144	<0.0001			443	<0.0001
0–23	3.0 ^a	0.1			5.2 ^a	0.2		
23–38	0.5 ^b	0.1			2.5 ^b	0.3		
38–50	0.3 ^b	0.1			1.6 ^c	0.2		

(a–c) Mean macroporosity values between three depths followed by a different letter differ statistically significantly ($P < 0.01$).

* Pore size at 0–23 cm layer >300 μm , at 23–38 cm layer >400 μm , and at 38–50 cm layer >550 μm .

Table 2

Effects of distance from a drain, and effect of depth on the number and area of earthworm burrows and on the number of root channels (Fig. 4). Degrees of freedom of the *F*-test (numerator, denominator), and statistical significances (*P*-values) are listed.

	Degrees of freedom		Number of earthworm burrows		Area of earthworm burrows		Number of root channels	
			<i>F</i> -ratio	<i>P</i> -value	<i>F</i> -ratio	<i>P</i> -value	<i>F</i> -ratio	<i>P</i> -value
Distance	2	6	3.4	0.10	3.2	0.11	0.1	0.95
Depth	2	6	58.9	<0.001	28.9	<0.001	13.2	0.01
Distance × depth	4	12	1.5	0.27	2.8	0.07	3.2	0.05

Fig. 4e). The differences between the soil layers were similar at each distance, the highest value being at 38 cm and the lowest at 23 cm depth (Fig. 4f). However, the magnitudes of differences in the number of root channels between three soil layers (depth) varied somewhat between distances ($P=0.05$ for distance-by-depth interaction, Table 2).

The mean number and area of earthworm burrows decreased as the distance from the drain line increased (Fig. 4a and c). The number of burrows was considerably greater above the drain than at the midpoint between two drains (44%, 89 burrows m^{-2} , $P=0.04$).

Further, the mean area of burrows was greater above the drain line than at the other locations at all depths ($P=0.11$ for the main effect of distance, Fig. 4a). The differences between the locations were greatest at 38 cm depth: 0.05 m^2 100 m^{-2} for the locations above a drain and 2 m from a drain, and 0.08 m^2 100 m^{-2} for the locations above a drain and 6–7 m from a drain ($P=0.07$ for distance-by-depth interaction).

Compared with the total volume of macropores (Table 1), the area of earthworm burrows, estimated as volume of earthworm burrows, was small, less than 0.3 m^2 100 m^{-2} (Fig. 4c and d). This was in agreement with earlier results for fine textured soils (Alakukku, 1996; Pitkänen and Nuutinen, 1997). Despite the small volume, continuous earthworm burrows may have a marked effect on saturated hydraulic conductivity (Schrader et al., 1995; Pitkänen and Nuutinen, 1998; Shipitalo et al., 2004), and water infiltration in massive subsoil (Ehlers, 1975).

Soil macroporosity (Table 1) and the number and area of earthworm burrows (Fig. 4b and d) decreased markedly as the sampling depth increased. This was in agreement with a number of previous results for macroporosity (Aura, 1995, 1999; Turtola and Paajanen, 1995; Olsen and Børresen, 1997; Øygarden et al., 1997) and earthworm burrows (Pitkänen and Nuutinen, 1995; Alakukku, 1996, 1997) in Finnish and Scandinavian arable clays.

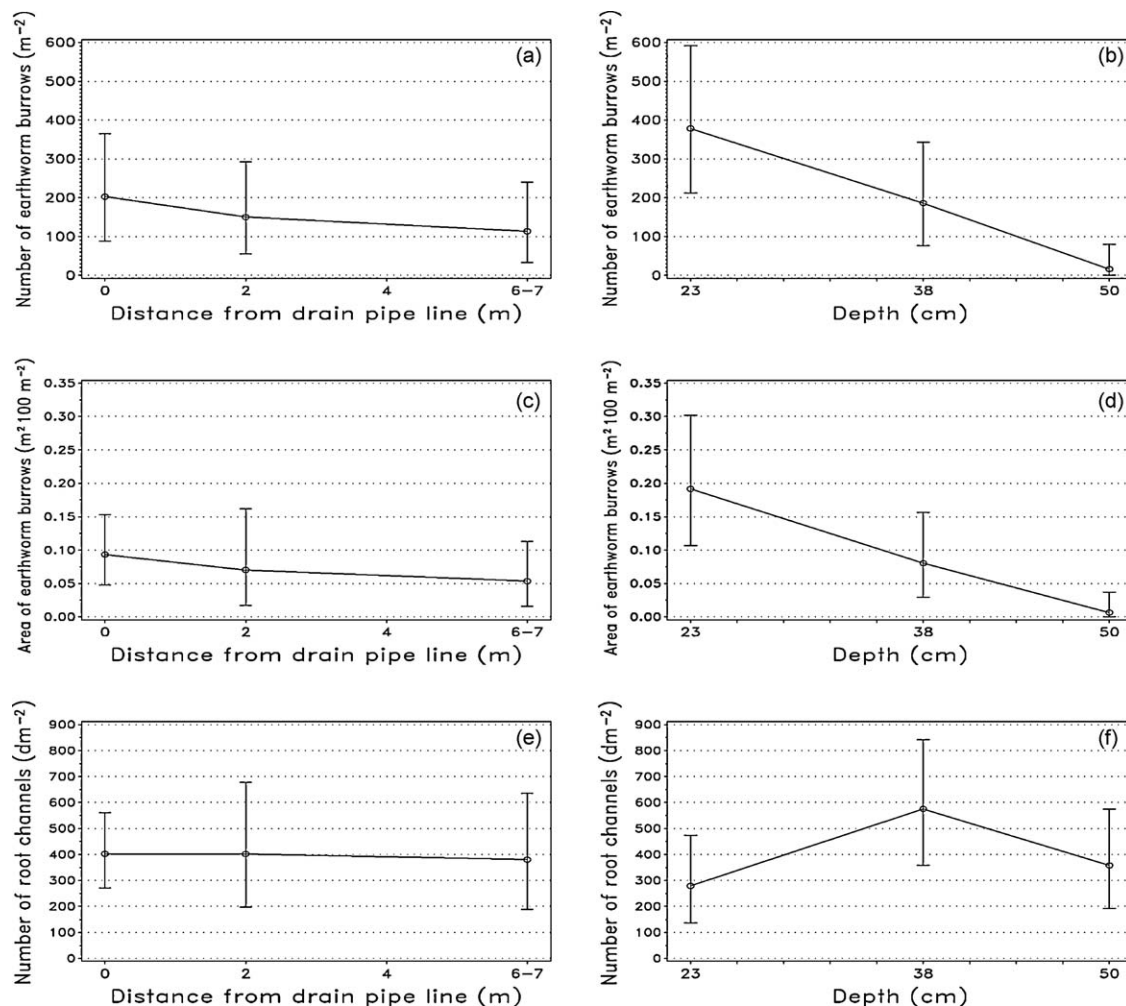


Fig. 4. Estimated means of number of earthworm burrows (a and b) and area of earthworm burrows (c and d), and number of root channels (e and f) at three depths and three distances in relation to the subsurface drain. Bars denote the 95% confidence intervals for the means. Results of statistical analyses are given in Table 2.

Table 3

Model-based mean estimates of soil saturated hydraulic conductivity measured at field (K_{fsat}) and in laboratory (K_{sat}) at three distances in relation to the subsurface drain. K_{sat} was determined at three depths. The means of K_{sat} are back-transformed to the original scale. CI is the 95% confidence interval of the mean. F as presented in Table 1.

	K_{fsat} (cm h ⁻¹)		$F_{2,2}$	P -value	K_{sat} (cm h ⁻¹)		$F_{2,6}$	P -value
	Mean	95% CI			Mean	95% CI		
Distance from drain (m)								
0	6.4 ^{ab}	2.3–10.6	278.61	0.004	5.5	1.7–11.5	0.97	0.43
2	4.4 ^b	1.9–7.0			7.2	2.7–14.0		
6–7	6.4 ^a	3.4–9.4			3.6	0.7–8.6		
Depth (cm)								
0–23	–	–			15.2 ^a	8.3–24.0	18.52	0.003
23–38	–	–			5.7 ^b	1.9–11.5		
38–50	–	–			0.4 ^c	0.1–2.7		

(a–c) Mean K_{fsat} or K_{sat} values between three distances or depths followed by a different letter differ statistically significantly ($P < 0.05$).

3.2. Saturated hydraulic conductivity

Saturated hydraulic conductivity of the soil, determined on the field (K_{fsat}) and in the laboratory (K_{sat}), relative to the distance from subsurface drains is shown in Table 3. Unexpectedly, the mean K_{fsat} was 31% lower at 2 m from the drain line than on or at 6–7 m distance. There were no marked differences in the K_{sat} values in relation to drain position. As the depth increased, the mean K_{sat} values decreased rapidly, from 5 to less than 0.5 cm h⁻¹ (Table 3).

3.3. Earthworms

The earthworm community of the field was strongly dominated by endogeic species. Of the 867 individuals counted, 95% belonged to these topsoil burrowing species (*Aporrectodea caliginosa* Sav., 36% of the individuals; *Aporrectodea rosea* Sav. 1%; *Aporrectodea juveniles* 58%). Of the deep burrowing, anecic, earthworm species, *L. terrestris* L. (2%) was found locally, mostly towards the eastern margin and in downslope positions of the field section (Fig. 5a). Of the litter-dwelling, epigeic, group, only *Lumbricus rubellus* Hoffm. was present but the number of counted individuals was low (<1%). Other individuals were *Lumbricus* juveniles and unidentified specimens.

The mean total number of earthworms was somewhat greater above the subsurface drains than between the drains (127 individuals m⁻² vs. 109 individuals m⁻², SE = 30), but the difference was not statistically significant ($P = 0.30$). The result for the mean fresh mass of earthworms was similar (25.2 g m⁻² vs. 18.1 g m⁻², SE = 8.1, $P = 0.44$). The western most sample pair of each drain line differed from the rest, with earthworm numbers higher between the drains than above the drain (Fig. 5b). When these sample pairs were disregarded, the earthworm total number was on average 58 individuals m⁻² higher above the drains (SE = 12.0, $P = 0.02$). Also with this adjustment, the mean fresh mass was twice as high above the drains, but the difference remained statistically non-significant ($P = 0.19$).

L. terrestris was found at seven of the 15 sampling sites. Only at one site was it present between the drain lines; in all other cases it was found in the vicinity of the drains (above or 2 m away). This finding is in accordance with an earlier observation of raised *L. terrestris* densities near drains, possibly due to the lowered groundwater table (Nuutinen et al., 2001; Nuutinen and Butt, 2003). The fact that the total abundance of earthworms at the Sjöckulla site was not consistently higher above the drains may be explained by the dominance of shallow working species. Their distribution is likely to be less directly affected by the depth of the groundwater table.

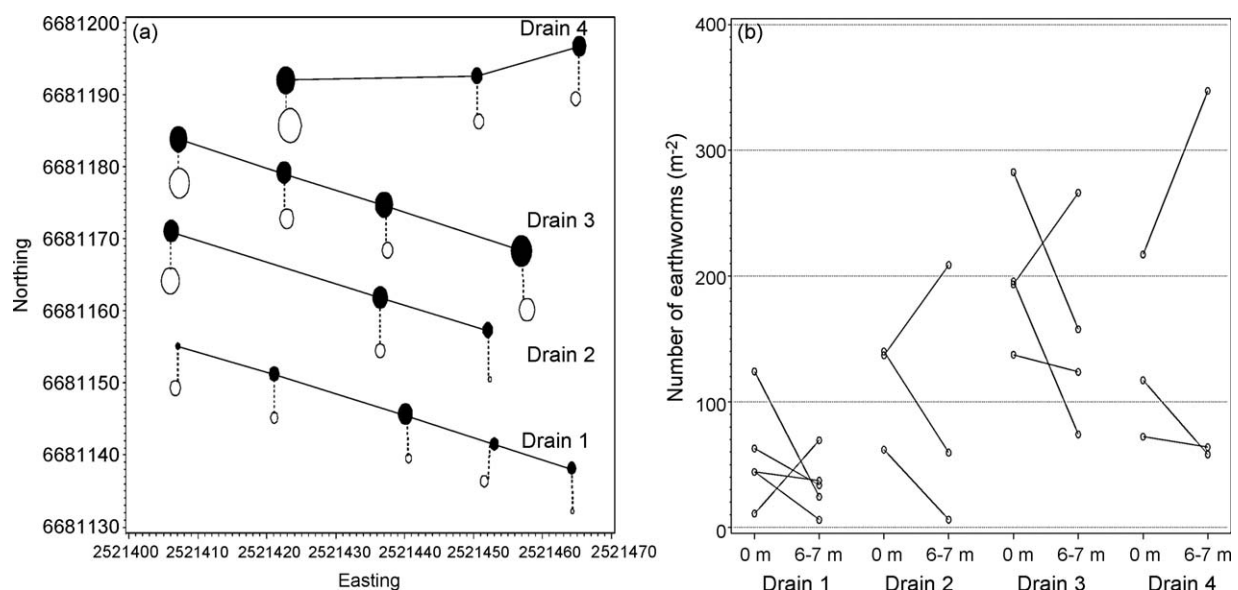


Fig. 5. (a) Spatial distribution of the number of earthworms. Filled circle (●) denote points above drain and open circles (○) points between two drains (6–7 m from drain). The size of the circle increases with the number of earthworms (range 6–347 earthworms m⁻²). (b) Number of earthworms at each sampling site along each of the four drain lines. Lines connect the observations of each distance pair. The symbol (0 m) in x-axis refers to a location above a drain, and the symbol (6–7) refers to a midpoint between two drains.

Table 4

Variation of soil properties along the slope. Medians (Md) and quartile deviations (QD) of soil physical properties, and the number and mass of earthworms at 6–7 m distance from a subsurface drain. K_{fsat} is saturated hydraulic conductivity measured in the field, K_{sat} is saturated hydraulic conductivity measured in the laboratory. Location of the drains is shown in Fig. 1.

Determination ^a	Drain 1		Drain 2		Drain 3		Drain 4	
Depth (cm)	Md ^b	QD	Md	QD	Md	QD	Md	QD
Macroporosity >30 μm ($\text{m}^3 100 \text{ m}^{-3}$)								
0–23	5.2	0.2	6.1	0.3	4.6	0.1	5.7	0.3
23–38	2.7	1.3	2.1	0.3	2.4	0.3	3.4	0.1
38–50	1.7	0.4	1.4	0.2	1.4	0.1	2.3	0.2
Earthworm burrows m^{-2}								
23	170	28	340	57	594	28	509	6
38	0	28	113	113	283	170	57	19
50	0		0	14	0	7	0	9
Root channels m^{-2}								
23	195	32	225	39	415	41	305	66
38	390	120	388	51	1113	59	585	168
50	190	27	225	57	370	146	415	85
K_{fsat} (cm h^{-1})	5.6	0.7	6.9	1.4	7.5	1.1	4.6	2.9
K_{sat} (cm h^{-1})								
0–23	4.0	1.2	8.7	25.1	10.6	4.5	43.8	2.3
23–38	0.03	0.05	0.01	1.5	3.1	5.4	1.0	7.4
38–50	0.005	0.001	0.004	0.001	0.006	0.03	0.006	0.01
Earthworms m^{-2}	34	6.4	60	51.0	141	63.1	64	72.3
Earthworms (g m^{-2})	8.2	4.4	9.9	5.4	27.1	4.6	28.9	10.1

^a Limits recommended for the good structure of clay soils in Finland: macroporosity (>0.03 mm) in topsoil $\geq 10 \text{ m}^3 100 \text{ m}^{-3}$ (Czeratzki, 1966); K_{sat} in topsoil $\geq 30 \text{ cm h}^{-1}$, in subsoil $\geq 1 \text{ cm h}^{-1}$ (Aura, 1990); number of earthworm burrows in subsoil $\geq 100 \text{ m}^{-2}$ (Aura, 1990; Alakukku, 1996); number of root channels in subsoil $\geq 300 \text{ dm}^{-2}$ (Alakukku, 1996; Aura, 1999).

^b $n = 5, 4, 4, 3$, for drains 1, 2, 3, 4, respectively.

3.4. Soil properties and earthworm abundance in relation to slope

The slope of the experimental field was an important source of variation for many soil properties and for earthworm abundance and was taken into account in statistical modelling by including the drain and distance as an explanatory variables in the models. When the locations midway between two drains (6–7 m from a drain) were considered, the median values of biopores, K_{sat} , and the number and mass of earthworms tended to be smaller at the upper (drain 1) than the bottom part of the slope (drain 4) (Table 4). A similar trend was seen in bioporosity and K_{sat} measurements above the drain and at 2 m from the drain. For instance, at a depth of 50 cm, no earthworm burrows was observed above the drain 1 (median $K_{sat} < 0.0001 \text{ cm h}^{-1}$ in the layer of 38–50 cm) but above drain 4 the median number of earthworm burrows was 170 m^{-2} ($K_{sat} 0.02 \text{ cm h}^{-1}$; rest of the data not shown).

In the layer 23–38 cm above the drain at the bottom part (drains 3 and 4) of the slope, the median K_{sat} value was greater than 1 cm h^{-1} and the number of earthworm burrows was more than 100 m^{-2} (data not shown). These are the recommended lower limits for good soil structure in Finnish conditions (Table 4). At all distances from the drain line in the bottom part of the slope (drains 3 and 4), the median number of root channels exceeded the value of 300 dm^{-2} , which is likewise regarded as a lower limit for favourable soil structure (Table 4).

Good soil structure at the bottom part of the slope was slightly unexpected since at least in the sampling year the groundwater level was higher at the bottom than in the upper part of the slope (Fig. 3). During the growing season, however, the groundwater level at the downslope was close to the depth of the subsurface drains, and the differences in soil moisture content in the layer 0–60 cm were relatively small between slope positions (Fig. 3). The mean clay content of the soil below 25 cm depth was clearly less at the bottom part of the slope (drain 4) than at the top (drain 1, Fig. 2). This may have been one reason for the greater earthworm abundance and the higher values of biopores and K_{sat} at the

bottom part. In arable fields, high clay content of soil, such as that found in the upslope part of the field, is often related to soil compactness, which may be detrimental for earthworms (Bostrom, 1986).

The vertical distribution of the clay content varied along the slope which may be partly explained by geological reason but partly also by water and tillage erosion. Ploughing and surface runoff parallel to the slope may have entrained and transferred topsoil material downslope.

3.5. General discussion and conclusions

The primary goals of agricultural drainage in humid regions are to provide site trafficability for timely sowing and harvesting, and to lower the water content of the root zone for adequate aeration after excessive snowmelt/rainfall. Drainage is also important to the structure development of clay soils because unsaturated conditions are favourable for physical and biological processes that enhance formation of soil aggregates and macropores. The effect of subsurface drainage on soil properties has been investigated by comparing drained and undrained (only ditches around the field) soils (Hundal et al., 1976; van de Graaff, 1979; Bouma et al., 1979). Our approach was to study soil structure and make interpretations about water movement in relation to subsurface drain position within a single field section as all the fields in our study region were subsurface drained.

At depths ranging from 0 to 50 cm below soil surface, the volume of soil macropores, the number of root channels, and the saturated hydraulic conductivity (K_{sat}) did not differ markedly with the horizontal distance from the subsurface drain line. These results are in agreement with the findings of Fausey and Lal (1989) who report that water infiltration into silty clay soil was not significantly affected by the distance from a subsurface drain about 30 years after drainage. In contrast to this, Øygarden et al. (1997) found that K_{sat} of a clay soil (clay content 25–45 $\text{g } 100 \text{ g}^{-1}$) at the depth of 20–60 cm was clearly greater above the subsurface drain

than at the midpoint between drains two or three years after the drains were installed. The trench had been backfilled with the excavated soil. Messing and Wesström (2006) found that on clayey soils field saturated steady state infiltration and K_{sat} were greater in trench backfill zone than at the midway between trench lines in plough sole and subsoil even 45 years after the installation of subsurface drains.

Comparison against literature values reveals that for the whole sloping field section, the K_{sat} and macroporosity of the subsoil were clearly greater than the values for poorly drained, heavy clay soil that was not sloped (Turtola and Paajanen, 1995; clay content 83–90 g 100 g⁻¹). In the present study, the variation of groundwater level in relation to distance from subsurface drain line was less than was expected. According to measurements during the growing season, the depth of groundwater table at the midpoint between two drains was frequently about the same or deeper than the installation depth of the drains. Corresponding to the behaviour of the groundwater table, the differences in soil moisture content (0–30 and 0–60 cm depths) were small between different distances from subsurface drain line. Groundwater level and soil moisture determinations indicate that the drying history of soil at different horizontal distances from drain line did not vary markedly explaining the non-existing horizontal gradient in determined soil properties.

When drain tiles are installed, soil layers are mixed during the digging, and the structure of the soil excavated from the trench and used again to fill the trench is thoroughly disturbed. Aura (1990) found that K_{sat} for soil used as filling material for the trench was greater than that of undisturbed soil. After the drains were installed the loose backfill material was evidently compacted naturally by soil water movement and weight of the overlying soil. For instance, the differences in dry bulk densities in the soil layer extending 50 cm below the soil surface were relatively small between distances from a drain (Alakukku, unpublished data). We can presume that the compression process decreased the difference in macroporosity between the disturbed backfill soil and the original soil. Vakkilainen (1980) found that both macroporosity and infiltration into a trench decreased noticeably during the first four to five years after the drain installation as the soil was compacted by natural forces and field traffic. Since subsurface drains were installed in our field 50 years before the time of soil sampling, we assume that the soil structure above the drains had largely regenerated.

The subsurface system had been beneficial for the formation of soil porosity owing to its effects on earthworm activity. The number of earthworm burrows was clearly greater above the drain than at midpoint between two drains. Correspondingly, except for the western part of the field, where the depth of drainage was shallowest, the number of earthworms was greater above drain than midway between drains.

Our results diverge from those reported by Shipitalo et al. (2004). They determined saturated hydraulic conductivity of a Finnish clay soil (40–46 g 100 g⁻¹ clay) on a grass field (K_{fsat}) 50 years after the installation of subsurface drains, and found that the mean K_{fsat} above subsurface drains was twice as high as at the midpoint between the drain lines. Direct contacts between the burrows of *L. terrestris* and subsurface tiles were evidently one reason for the high K_{fsat} above the drain lines. The characteristics of the earthworm community at the Sjöckulla field together with the observations on depth distribution of the earthworm burrows suggest that such direct burrow–tile contacts were not common in our field. Earlier findings from Finnish arable clay suggested that during the growing season, the activity of *A. caliginosa*, the dominant species in the present field, would mainly be limited to the top 0–30 cm soil layer (Pitkänen and Nuutinen, 1998). Burrowing beyond that depth is likely to occur, for instance,

when individuals flee frost, but burrowing down to the subsurface drain depth has not been observed.

In areas where *L. terrestris* was present, its population density remained relatively low, from 2 to 10 individuals m⁻². In an earlier study, a correspondingly low average density had a notable impact on soil macroporosity and hydraulic conductivity owing to the large numbers of old, unoccupied burrows in the subsoil (Shipitalo et al., 2004). It is possible therefore, that, inside the Sjöckulla field, the areas inhabited by *L. terrestris* delimit zones with high rate of by-pass flow into the subsurface drains.

The largest macroporosity (classified here pores diameter > 300 µm) is expected to have a major effect on subsurface water flow and nutrient transport on the experimental field. Compelling evidence for the preferential transport of water, nutrients and suspended soil into the subsurface drains was obtained in the experimental field. For instance, subsurface drain flow was observed only a few hours after a rainfall event even in late autumn, indicating high soil hydraulic conductivity and functional macroporosity in the swollen clay soil (Paasonen-Kivekäs et al., 1999). According to our results, the preferential flowpaths were likely the earthworm burrows and biggest root channels. Cracks were other probable flowpaths, especially at the bottom part of the slope where the soil cracking was noticeable during the growing season until early autumn. Although the spatial variation of soil structure was high throughout the field, preferential flow was more likely to occur at the bottom of the slope, where the subsoil had more biopores and higher saturated hydraulic conductivity, than in the upper part of the slope.

So-called dual porosity models (coupling macropore and micropore flow) are commonly used in describing the non-uniform flow in macroporous soil (e.g. Beven and German, 1981; Jarvis and Larsson, 2001). Hintikka et al. (2008) modelled the effects of macroporosity on runoff processes in the study field using the present results in the parameterisation of one-dimensional, dual porosity model MACRO model (Jarvis and Larsson, 2001). The vertical distribution of macropore volume (>0.3 mm) and the distribution of saturated macropore hydraulic conductivity were according to the average values of the measurements. The results of the modelling study suggested that most of the subsurface drain flow was generated through the macropore domain. It was suggested that use of a two- or three-dimensional modeling approach becomes advisable when prediction of proportional fractions of drain flow and surface runoff, and simulation of groundwater level in a sloping field are of interest (Hintikka et al., 2008).

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